

JARED Technical Architecture Specification

Version 0.1 – Local-Only Platform Design

1. System Overview

- Local-only voice assistant and device automation platform.
- Modular architecture with strict separation of concerns.
- All network traffic gated by central NetPolicy module.
- Device control executed only through Policy Engine.
- Scalable from single-room to multi-room deployment.

2. Core Architecture Layers

2.1 Audio Layer

- Audio capture pipeline (mic input).
- Voice Activity Detection (VAD).
- Wake word engine abstraction.
- Local Speech-to-Text engine wrapper.
- Text-to-Speech rendering + mixer.
- Music ducking and barge-in controller.

2.2 NLP & Dialogue Layer

- Intent schema and slot-based entity model.
- Rule-based classifier (v0.1).
- Dialogue planner for multi-turn context.
- ContextFrame per room session.
- Decision capture and troubleshooting extraction.

2.3 Knowledge Layer

- Manual ingestion pipeline (PDF text + OCR).
- Keyword index (semantic index later).
- Retrieval engine returning PageChunks.
- Answer composer (procedure formatting).
- Hidden provenance for on-demand source lookup.

2.4 Device Platform

- Device registry stored in SQLite.
- Capability-based device templates.
- ActionRequest abstraction.
- Policy Engine enforcement (Class 0/1/2).
- Driver framework (MQTT first).
- Execution + verification module.

2.5 Security & Network Control

- Central NetPolicy controlling outbound traffic.
- Offline mode default.
- PIN phrase authentication system.
- Authorization window logic.
- Audit log for critical actions.

3. Core Data Models

3.1 ActionRequest

```
ActionRequest:
  request_id: UUID
  room_id: str
  device_id: str
  action: str
  params: dict
  invocation:
    trigger: wakeword | ptt
    user_role: owner | guest
  auth:
    pin_verified: bool
    ptt_active: bool
```

3.2 PolicyDecision

```
PolicyDecision:
  allowed: bool
  reason: str
  requires_auth: bool
  confirm_style: silent | brief | verbose
  verify_required: bool
```

3.3 DecisionCandidate

```
DecisionCandidate:
  id: UUID
  project_id: str
  summary: str
  details: dict
  confidence: float
  status: queued | approved | rejected | expired
```

3.4 TroubleshootingCard

```
TroubleshootingCard:  
  id: UUID  
  title: str  
  symptom: str  
  fix_steps: list  
  context: dict  
  verification: str
```

4. Device Safety Policy

- Class 0 – Safe: No auth required.
- Class 1 – Caution: Optional confirmation.
- Class 2 – Critical: Auth required for ON only.
- Class 2 OFF: No auth, but verified state confirmation required.

5. Knowledge Routing Policy

- Project procedural questions → Manuals first.
- Project decisions → Project memory store.
- Definitions → Offline reference pack.
- Online search only with explicit permission.
- Manual content never uploaded externally.

6. Local-Only Network Contract

```
All outbound HTTP calls must pass through NetPolicy.  
If NetPolicy.allow_internet == False:  
    block request  
    log attempt  
    notify system
```

7. Scalability & Future Extensions

- Multi-room mic arbitration via RoomResolver.
- Voice identity recognition layer.
- Automation rules engine (trigger/condition/action).
- Plugin/pack marketplace architecture.
- Local mobile interface (LAN-only API).

8. Versioning & Upgrades

- SQLite migrations folder for schema evolution.
- Config schema version control.
- Updater module with rollback support.

- Diagnostics bundle export for debugging.